What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Disseminated Intravascular Coagulation

Last updated October 8, 2025 · 12 min read

Adapted by: Chelsey Deel

Editor: Jordan Finn

Adapted from: Disseminated Intravascular Coagulation

Listen to this Brick

0:00 / 21:23 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of DIC.
2. Explain the pathophysiology of DIC.
3. Interpret clinical findings to properly diagnose DIC.
4. Outline the

CASE CONNECTION

LH is a 22-year-old college senior who is brought to the emergency department via ambulance. LH's roommate is at his bedside. "He's been feeling sick for the past 2 days," his roommate reports. "This morning he had a severe headache and neck pain. He seemed groggy. Then he collapsed on the couch, and I couldn't wake him up." LH is currently unresponsive. His vitals are a temperature of 103.4°F, heart rate is 126/min, respiratory rate is 22/min, and blood pressure is 80/60 mm Hg. You note that his neck is rigid when you raise him up in bed to listen to his lungs. He has diffuse petechiae, and LH didn't stop bleeding at the puncture site when an IV was started.

What will be your next steps in LH's evaluation and treatment? Consider your answers as you read, and we'll revisit LH at the end of the brick.

GO TO CONCLUSION

What Is Disseminated Intravascular Coagulation?

Disseminated intravascular coagulation (DIC) is a **thrombo-hemorrhagic disorder** (combining both clotting and bleeding) that involves the entire body and is typically seen in very sick patients, often in the intensive care unit. It's also considered a secondary form of **thrombotic microangiopathy**, a disease due to small thrombi occluding multiple small blood vessels.

If you can remember one thing about DIC, remember that it's not a primary disorder. Instead, it's **underlying severe medical disease**. So, if you have a patient with DIC, search for the underlying cause to stop the DIC.

Let's begin by examining the name: "disseminated" means that the condition is systemic or widespread, "intravascular" refers to the process occurring within blood vessels, and "coagulation" indicates that blood clotting is involved. Widespread blood clotting within the blood vessels can use up platelets and coagulation factors, leading to bleeding.

How Do Patients with Disseminated Intravascular Coagulation Present (LO #1)?

Because DIC has so many different causes, the clinical presentation varies widely. The onset may be acute (as in sepsis) or more insidious (as in cancer). Some cases are dominated by bleeding, and others present with extensive clotting (gangrene of fingers, toes, or skin; infarcts in internal organs).

DIC is often a **multi–organ system disease**, with potential to affect the lungs (causing cyanosis and respiratory failure), nervous system (e.g., seizures, coma), renal system (e.g., oliguria, acute renal failure), and cardiovascular system (e.g., circulatory failure, shock), to name a few. The patient may also show signs of bleeding from intravenous or other lines, and the skin may show **petechiae**, the microvascular hemorrhages typical of patients with low platelets due to any cause (Figure 1).

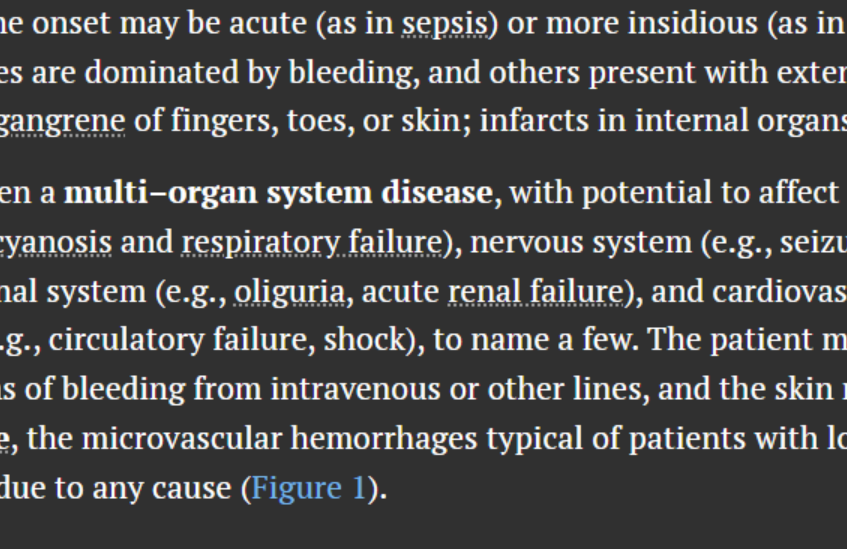

Figure 1

What is the most common skin manifestation of disseminated intravascular coagulation (DIC)?

What Is the Pathophysiology of Disseminated Intravascular Coagulation (LO #2)?

DIC begins when som plications triggers the coagulation cascade.

Etiology

The causes of DIC can be broken down into two categories: dumpers (things that dump procoagulants into the blood) and rippers (things that damage the endothelium).

Conditions that dump procoagulant substances into the blood include:

- Obstetric complications (e.g., abruption, amniotic fluid embolism)
- Adenocarcinoma (especially very mucinous ones, like pancreatic adenocarcinoma)
- Venomous snake bites
- Acute promyelocytic leukemia

Conditions that damage endothelium or tissue include:

- Sepsis, especially severe, gram-negative, toxin-producing bacterial infections
- Trauma or thermal burns
- Vasculitis (e.g., systemic lupus erythematosus)

Mnemonic

The r OST.

Thrombosis

How do these events initiate DIC? As stated, some cause release of a procoagulant substance (like tissue factor) into the circulation, while others cause endothelial injury to the blood vessel (Figure 2).

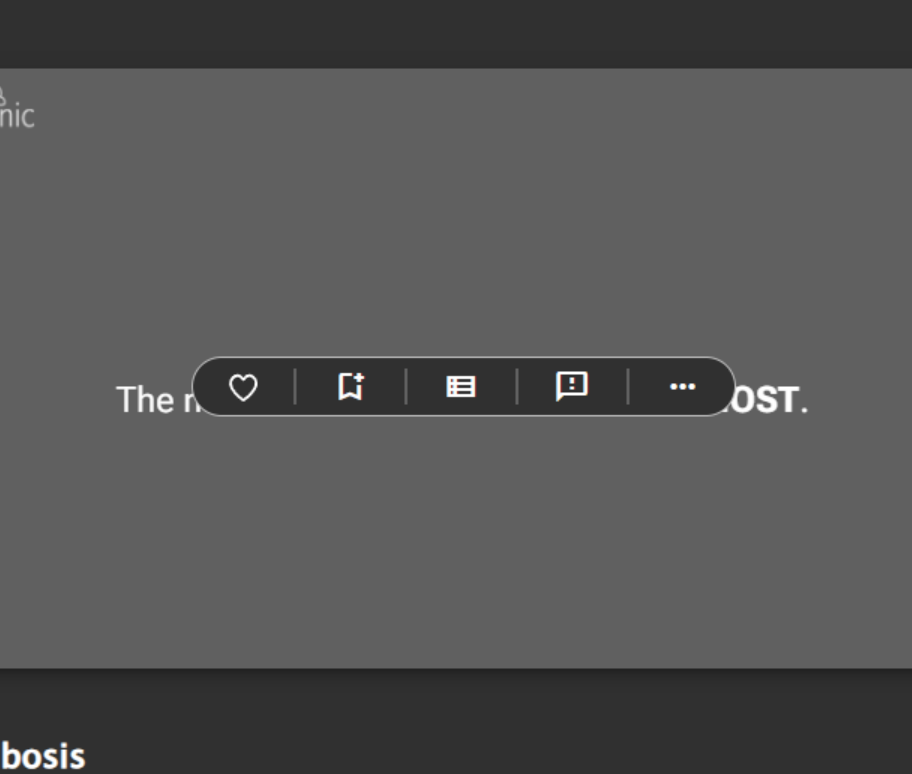

Figure 2

The vessel wall injury releases tumor necrosis factor (TNF), which causes endothelial cells to release tissue factor. This is the stimulus for the extrinsic pathway of the coagulation cascade to activate. So, the coagulation cascade is activated when it shouldn't be—and it's supercharged, making excessive amounts of unnecessary **fibrin** in vessels all over the body. All this fibrin lodges in small vessels and combines with platelets to make **thrombi**. So, DIC can be considered a secondary form of **thrombotic microangiopathy**, which has several other causes.

Thrombi can occlude vessels, which reduces regional blood flow, causing **tissue ischemia** (lack of oxygen) and sometimes tissue necrosis. They also damage passing red blood cells (RBCs), leading to anemia from hemolysis of the RBC, or **microangiopathic hemolytic anemia (MAHA)**. MAHA is characterized by abnormal schistocytes (Figure 3).

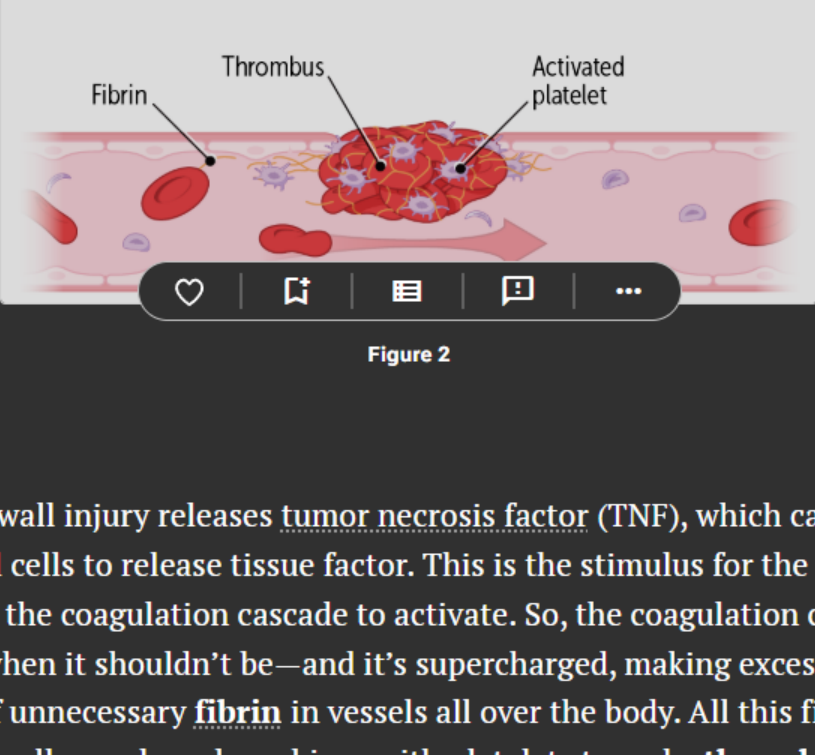

Figure 3

Hemorrhage

How does DIC also cause hemorrhage? All the rampant clotting in DIC consumes most of the available platelets and coagulation factors, which can lead to hemorrhage. To make matters worse, as the body tries to lyse the clots, fibrin is broken down into fibrin degradation products (which themselves inhibit clotting and aggravate the bleeding problem). That's how DIC can manifest with both systemic microthrombi and hemorrhage at the same time.

What initiates disseminated intravascular coagulation (DIC)?

Histopathology

On a peripheral blood smear, you may see schistocytes and reduced platelets. These are seen in any form of thrombotic microangiopathy, not just DIC.

Schistocytes. RBCs are very pliable, but they can't easily flex as they migrate through the forming clot, including its sharp strands of fibrin (see Figure 3). Some RBCs are destroyed completely, and others are sliced into two parts that reseal themselves.

These RBC fragments are smaller than normal RBCs and are angular rather than round. Figure 4 shows almost too many schistocytes to count.

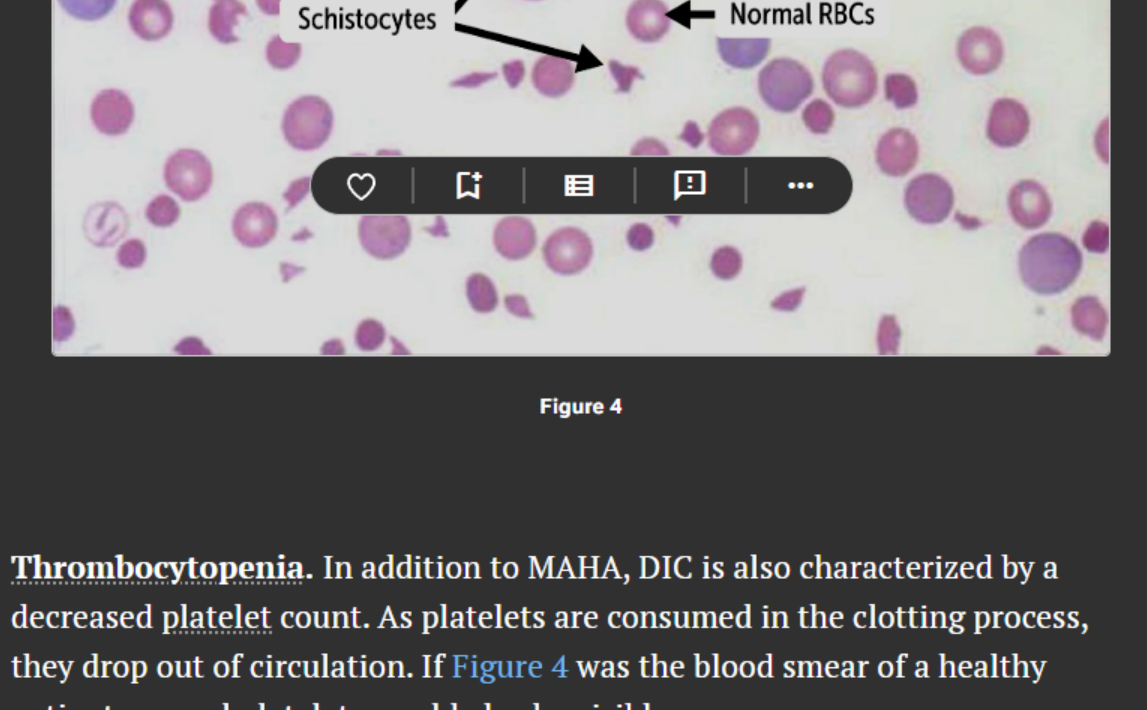

Figure 4

Thrombocytopenia. In addition to MAHA, DIC is also characterized by a decreased platelet count. As platelets are consumed in the clotting process, they drop out of circulation. If Figure 4 was the blood smear of a healthy patient, several platelets would also be visible.

Why do schistocytes appear in disseminated intravascular coagulation (DIC)?

How Do We Diagnose Disseminated Intravascular Coagulation (L

To make a diagnosis of DIC, perform a careful history and physical exam, looking for other causes of thrombotic microangiopathy such as drugs, infections, and malignancy.

Blood Smear

Characteristic blood smear findings include **MAHA** (see Figure 4) and a **low platelet count**.

Coagulation Tests

Because the main problem in DIC is rampant, uncontrolled fibrin production and clot formation, the coagulation factors will eventually be depleted, and the prothrombin time (PT), international normalized ratio (INR), activated partial thromboplastin time (aPTT), and thrombin time (TT) are all **prolonged**. This distinguishes DIC from the other forms of thrombotic microangiopathy.

Fibrin Degradation Products

You can also look for **elevated levels of fibrin degradation products (FDPs) or D-dimers** to see if the patient is breaking down clots (as would be expected in DIC). These form any time a clot starts to break down. However, because these tests are super-sensitive, they are elevated even in conditions of minor clot formation. So, they are not specific tests and are therefore not very helpful in proving that a patient has pathologic clotting. Although not specific, they are sensitive for diagnosing DIC (few false negatives) and are great at **ruling out** pathologic clotting. If the FDP level is low, you can be reasonably confident that a major clot isn't osis of DIC is made, monitoring FDP levels condition is improving with treatment.

Other Serum Tests

Typically, the serum indirect bilirubin is elevated due to the ongoing hemolysis. Fibrinogen is also typically decreased in DIC.

How can measuring the fibrin degradation product (FDP) level contribute to a diagnosis?

How Do We Treat Disseminated Intravascular Coagulation (LO #4)?

The first goal of managing DIC is to **treat the underlying condition.** In some patients, we may also need to more directly correct their coagulopathy, especially if they have significant bleeding or will undergo a medical or surgical procedure th or significant bleeding in DIC may include transfusion of some or all of the following:

- Platelets
- Red blood cells
- Cryoprecipitate to replace fibrinogen and factor VIII
- Fresh frozen plasma (FFP) to correct PT and aPTT.

If thrombosis is the predominant feature of the patient's DIC, treatment decisions must weigh the risk of bleeding with the risk of forming more thrombi. For large thrombi or tissue necrosis, anticoagulation with heparin is used.

How are blood-derived products used in the treatment of disseminated intravascular coagulation (DIC)?

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to LH and treatment?

You suspect LH has *Neisseria meningitidis* and sepsis and, as a result, disseminated intravascular coagulation. This is a medical emergency, so antibiotics are ordered as the diagnostic evaluation begins with a complete blood count, blood cultures, and coagulation studies. The coagulation studies show elevated prothrombin and activated partial thromboplastin times. LH's white blood cell count (WBC) is high, hemoglobin low, and platelet count very low. Schistocytes are reported on the peripheral smear. LH is transferred to the ICU where his abnormal coagulopathy is corrected. Results from a lumbar puncture come back positive for *N meningitidis*. Antibiotics are continued, and in 72 hours LH's hematologic picture improves. He continues to progress well during the remainder of his hospital stay and is discharged on day 7.

Summary

- Disseminated intravascular coagulation (DIC) is a syndrome involving aberrant activation of the clotting cascade, leading to widespread thrombus formation, consumption of coagulation factors and platelets, and bleeding.
- DIC is a thrombo-hemorrhagic disorder in which some cases are dominated by thrombotic complications and others by bleeding.
- Clinical manifestations can affect any organ system.
- Symptoms may include bleeding, tissue ischemia, tissue infarcts, and shock.
- Most cases of DIC are associated with MOST: malignancy, obstetric complications, sepsis, and trauma.
- Damage to endothelium or dumping of tissue factors into the blood sets off the coagulation c clot formation.
- The clot consumes platelets and causes damage to red blood cells (RBCs) (schistocytes).
- The blood smear shows schistocytes and a decreased platelet count.
- Prothrombin time, activated partial thromboplastin time, and thrombin times and international normalized ratio, are all prolonged/increased.
- Fibrin degradation product and D-dimer levels are increased, but this test is nonspecific for DIC.
- Treating the underlying condition is most important.
- Some patients require correction of the coagulopathy and anemia.
- If thrombosis is the predominant feature, treatment decisions must weigh the risk of bleeding with the risk of forming more thrombi.

Review Questions

1. **How do fibrin degradation products directly contribute to the abnormal bleeding seen in disseminated intravascular coagulation (DIC)?**

A. ○ Create new injury, exposing subendothelial collagen
B. ● Inhibit new clot formation
C. ○ Inhibit protein C formation
D. ○ Initiate widespread microvascular thrombosis
E. ○ Stimulate platelet aggregation

Hide Explanation

The correct answer is inhibit new clot formation (B). Fibrin degradation products (FDPs) are the breakdown products of the fibrin within blood clots. They are not just inert fragments, however; they actively prevent formation of new blood clots. Create new injury, exposing subendothelial collagen (A), inhibit protein C formation (C), initiate widespread microvascular thrombosis (D), and stimulate platelet aggregation (E) would all increase thrombosis, not bleeding.

2. **After a complicated, full-term delivery of her first child, a 23-year-old female had a large postpartum hemorrhage that has been difficult to control over the past 24 hours. Which sets of laboratory tests would be most indicative of disseminated intravascular coagulation (DIC) in this patient?**

A. ○ Platelets ↑, fibrinogen ↑, PT ↓, PTT ↓
B. ● Platelets ↓, fibrinogen ↓, PT ↑, PTT ↑
C. ○ Platelets ↓, fibrinogen ↓, PT ↓, PTT ↑
D. ○ Platelets ↑, fibrinogen ↑, PT ↑, PTT ↑
E. ○ Platelets ↓, fibrinogen ↑, PT ↑, PTT ↓

Hide Explanation

The correct answer is platelets ↓, fibrinogen ↓, PT ↑, PTT ↑ (B). In disseminated intravascular coagulation (DIC), rampant clotting leads to the depletion of platelets and coagulation factors, causing a decreased platelet count, decreased fibrinogen levels (fibrinogen being a coagulation factor), and prolonged coagulation tests such as the prothrombin time (PT) and partial thromboplastin time (PTT). The lab results platelets ↑, fibrinogen ↑, PT ↓, PTT ↓ (A); platelets ↓, fibrinogen ↓, PT ↓, PTT ↑ (C); platelets ↑, fibrinogen ↑, PT ↑, PTT ↑ (D); and platelets ↓, fibrinogen ↑, PT ↑, PTT ↓ (E) are not consistent with the diagnosis of DIC.

3. **Which findings on a peripheral blood smear most point to disseminated intravascular coagulation (DIC)?**

A. ○ Blast cells
B. ○ Decreased platelets
C. ○ Increased white blood cells
D. ○ Large red blood cells with a purplish cast
E. ● Red blood cells with pointed contours and no central pallor

Hide Explanation

The correct answer is red blood cells with pointed contours and no central pallor (E). This describes a schistocyte (fragmented red blood cell [RBC]), which is never seen in normal blood and ation for disseminated intravascular coagulation (DIC) and its cause. Blast cells (A) in the blood are most often seen in leukemias and myeloproliferative disorders. Although patients with DIC have decreased platelets (B), there are more common reasons for a low platelet count, and without the presence of schistocytes, common causes should be investigated first. Increased numbers of white blood cells (C) (WBC) may be seen in infection, which if severe, may cause DIC. However, an increase in white blood cells alone is not indicative of DIC. Large red blood cells with a purplish cast (D) are reticulocytes. These are seen in small numbers in normal blood smears and in larger numbers in hemolytic processes.

4. **A patient presents to the hospital with sepsis, DIC secondary to bacterial meningitis. Which of the following is the best treatment option to manage their DIC?**

A. ● Antibiotics
B. ○ Cryoprecipitate
C. ○ Fresh frozen plasma (FFP)
D. ○ Heparin
E. ○ Platelet transfusion

Hide Explanation

The correct answer is antibiotics (A). Since the patient's DIC is secondary to their underlying bacterial i n antibiotics is the fundamental treatment. Treatment with product transfusions should be individualized to the patient and used to correct significant bleeding, not correct the abnormal coagulation studies alone. Cryoprecipitate (B) can be used if the patient is bleeding and fibrinogen levels are low. FFP (C) can be used if the patient is bleeding and has a decrease in other coagulation factors. Platelet transfusion (E) can be used if the patient is bleeding and has significant thrombocytopenia. Heparin (D) is used if the patient has predominantly thrombotic manifestations.

How would you rate this Brick?

👍 👎

References

- Greer JP, et al, eds. *Wintrobe's Clinical Hematology*. 14th ed. Wolters Kluwer; 2018.
- https://onlinelibrary.wiley.com/doi/10.1111/bjh.17172
- https://www.uptodate.com/contents/evaluation-and-management-of-disseminated-intravascular-coagulation-dic-in-adults?search=dic&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
- Kitchens CS, Kessle ve *Hemostasis and Thrombosis*. 4th ed. Elsevier; 2019. search=dic&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
- Kitchens CS, Kessler CM, Konkle BA, et al, eds. *Consultative Hemostasis and Thrombosis*. 4th ed. Elsevier; 2019.
- Kumar V, Abbas AK, Aster JC. Hematopoietic and Lymphoid Systems. In: Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease*. 10th ed. Elsevier; 2021.
- Provan D, ed. *ABC of clinical haematology*. John Wiley & Sons, Incorporated; 2018.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of DIC.~~
02 ☑ ~~Explain the pathophysiology of DIC.~~
03 ☑ ~~Interpret clinical findings to properly diagnose DIC.~~
04 ☑ ~~Outline the management plan of DIC.~~

Objective Confidence: 4/4